Sentence Smith: Controllable Edits For Evaluating Text Embedding Models

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In a nutshell

Research Problem

- **Controlled, *transparent* text generation** is still an open goal
  - Modern LLM prompting is powerful but **opaque** and hard to control, and costly.
  - Earlier symbolic approaches were limited by poor quality
- **Text Embeddings**: We hardly know much about the linguistic information that's contained in them.

Contributions

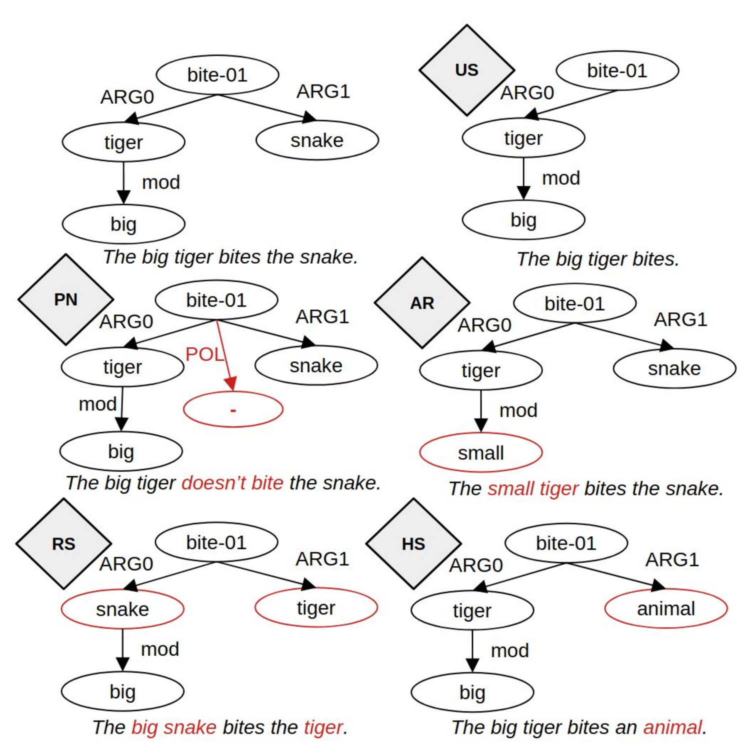
- Propose a resource-efficient **neuro-symbolic** framework (**Sentence Smith**) for transparent text edits
- **Application:** Use controllable edits to **generate hard negative (foil) pairs** for **fine-grained embedding evaluation**
- Allows **Dynamic Benchmark construction**
- Human validation shows generated texts have **good quality**

Goal: Generate Dynamic Embedding Benchmark

- Uses **Sentence Smith** to create foils, aka hard negatives
- Given a paraphrase, we apply a controlled manipulation
- Any embedding model should assign a higher similarity to the paraphrase than to the structurally similar non-paraphrase.

Interpretable Evaluation Categories

US: **U**nder**s**pecification
Remove information.
PN: **P**olarity **N**egation
Flip Polarity
AR: **A**ntonym **R**eplacement
Swap with antonym
RS: **R**ole **S**wap
Swap Semantic Roles
HS: **H**ypernym **S**ubstitution.
Swap with hypernym



Embedding Objective

sim(dissimilar structure & same meaning)
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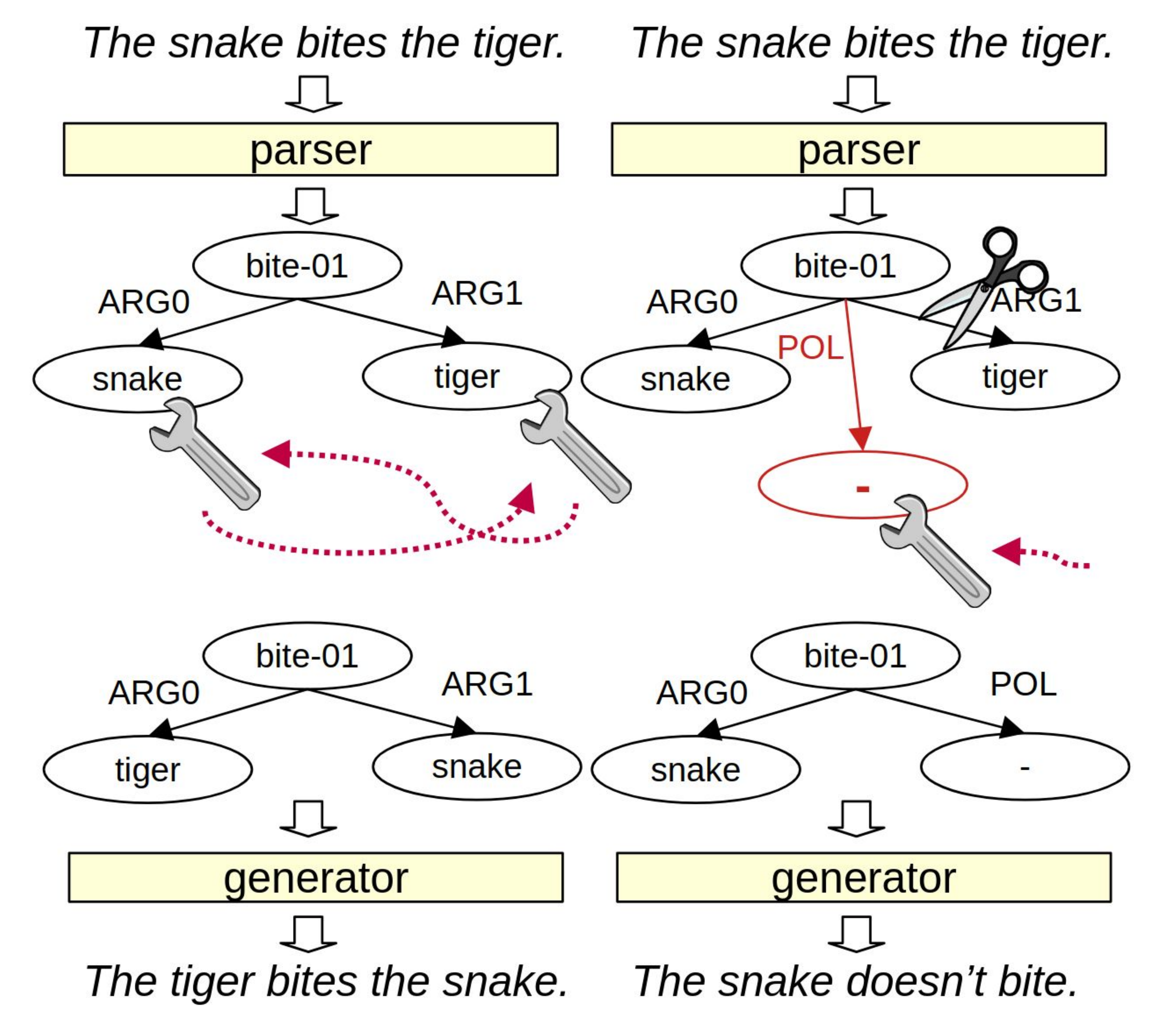
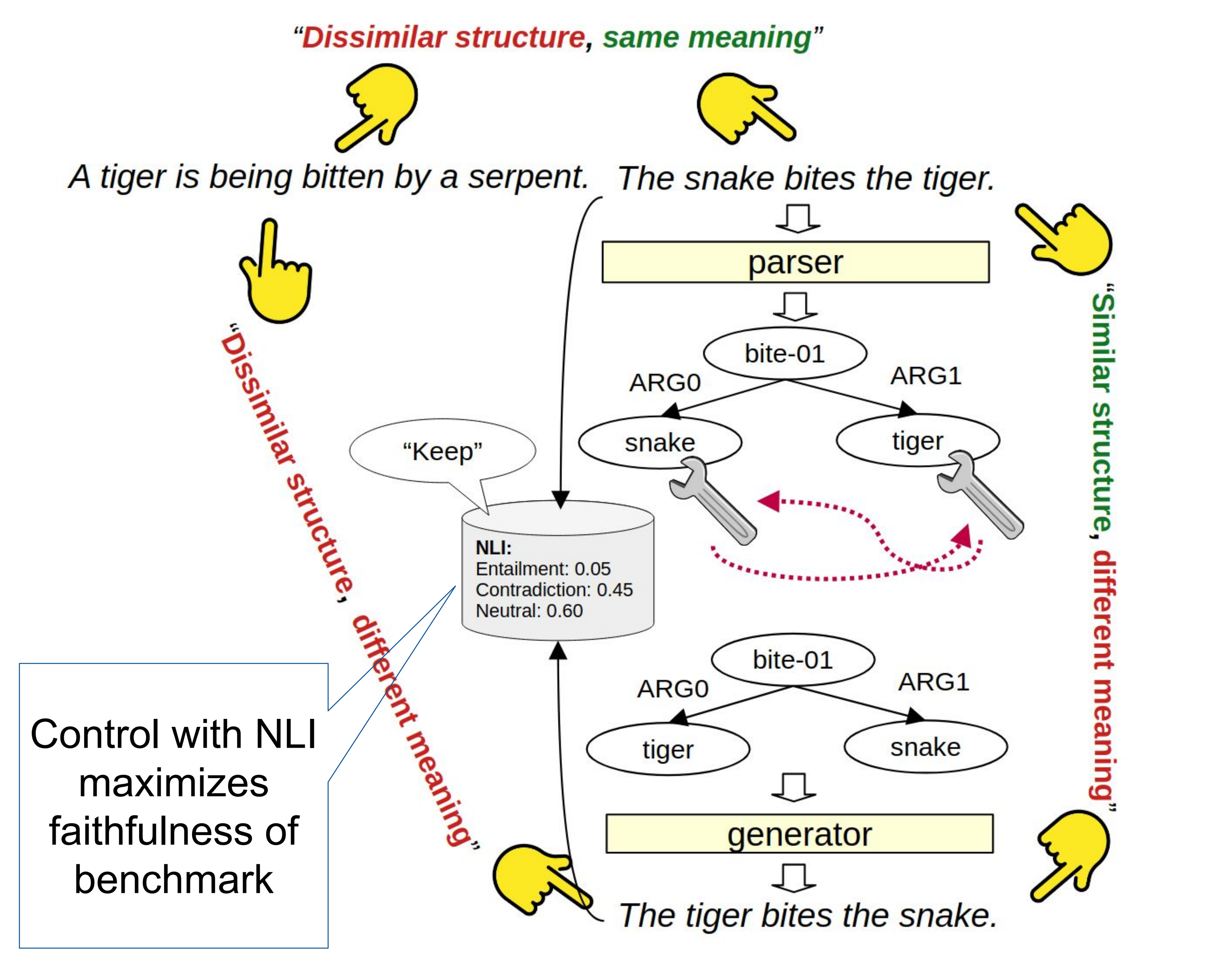
sim(similar structure & different meaning)

Sentence Smith Approach

- **Sentence Smith: PARSE → EDIT → GENERATE.**
- Uses efficient smaller parsing and generation models.





Embedding Benchmark Results